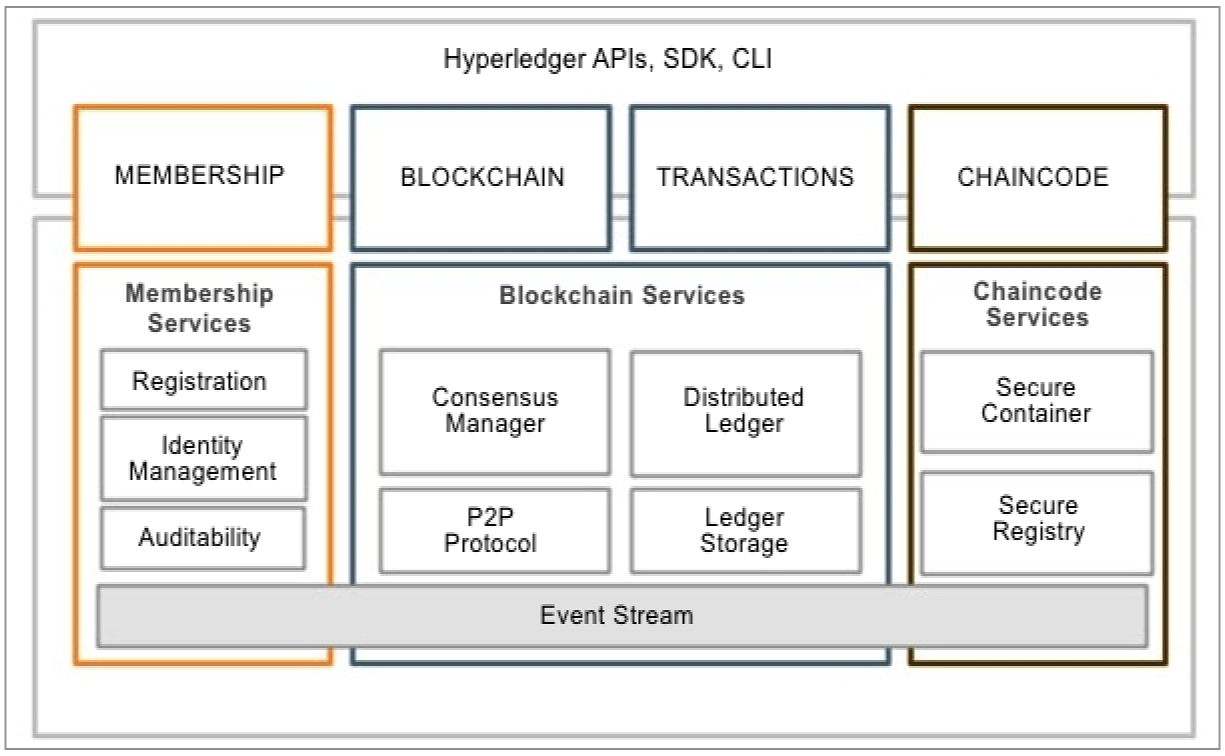

Fig. 2: Hyperledger services

and locks the transaction, a block is created, which contains encrypted information of the person who intends to transfer the money to B. It is created by combining a time-stamp, all significant information as well as a digital signature to ensure that the private key of A is attached to the public key of B to transfer the amount in the form of cryptocurrency.

Noding. After the secure encrypted block is created, it is broadcast as a message across the blockchain via all nodes attached to the network.

Nodes are computers connected to a single network. Once the block is communicated to the system, the nodes are primarily responsible for confirming each entry by solving an arithmetical problem connected to it to verify and validate the block and receive Bitcoin in return.

Verification. Once the transaction is confirmed, blocks are time-stamped and coupled together with previously verified blocks present in the blockchain. Once the process is complete, B receives the amount from A.

Open source blockchain platforms

Blockchain technology was first used for financial transactions, but now its scope is increasing rapidly. Today it is applied in a variety of industries like e-commerce, data management, energy, online voting, gaming, e-governance and more.

A number of commercial and open source platforms provide the framework for creating applications that support blockchain. A blockchain product must fulfil the following primary requirements:

- Create a live and distributed transactional database
- Create user identification labels for active parties
- Each transaction must be verified on an active ledger for approval
- Transactions should stop in case of any non-verification

To develop a smart mobile blockchain app, one needs to use a transaction and ledger model. For easy and quick blockchain app development, there are various open source platforms available, some of which are featured here.

Ethereum. Ethereum is an open source, public, blockchain-based decentralised platform running smart contracts—applications that run exactly as programmed without any possibility of downtime, censorship, fraud or third-party interference. It was proposed in 2013 by Vitalik Buterin and was crowdfunded online between July and August 2014.

Ethereum is a distributed public blockchain network. It is different from Bitcoin in terms of capability and purpose. Bitcoin blockchain is used to track ownership of digital currency, while Ethereum blockchain mostly focuses on running the programming code of any decentralised application.

In Ethereum, Ether is a crypto type, which is the heart of the network. It is used by application developers to pay transaction fees and services on Ethereum network.

Ethereum is made up of the following components:

- Ethereum virtual machine (EVM) is a 256-bit register stack that is completely sandboxed and isolated from the network, file system or other processes. Every node runs an EVM and executes the same instructions.
- Smart contracts are high-level programming abstractions compiled to EVM byte code and deployed to Ethereum blockchain for execution. These are programmed using various languages like Solidity, Serpent, LLL and Mutan.
- Ethereum has almost 50 per cent of the market share and more than 250 live decentralised applications (DApps), which are used for digital signatures, stock market predictions, gaming, social media and other applications.
- It makes use of Merkle trees to improve scalability and optimise transaction hashing.

Official website: *www.ethereum.org*

Latest version: 0.10.0

Hyperledger. Hyperledger is an open source collaborative effort hosted by Linux Foundation. It does not support Bitcoin or any other cryptocurrency. The objective of Hyperledger is to advance cross-industry collaboration by developing blockchain and distributed ledgers, with a focus on improving performance and reliability of these systems as compared to traditional cryptographic designs—to make these capable of supporting global business transactions by major technology, finance and supply chain companies. Fig. 2 highlights Hyperledger reference architecture.

The following are some Hyperledger frameworks:

- Hyperledger Sawtooth is a modular blockchain suite developed by Intel, which makes use of the consensus algorithm called Proof of Elapsed Time (PoeT).
- Hyperledger Iroha is a project of some Japanese companies to create an easy-to-implement framework for blockchain.